

Jingfeng Pan

Phone: (+1) 3122870395 | Email: jfpan01@uchicago.edu

Website: <https://jingfengpan.github.io/> | LinkedIn: www.linkedin.com/in/jingfeng-pan-a8b462264

EDUCATION BACKGROUND

University of Chicago, Physical Sciences Division

Chicago, IL

MPCS Program

Sep.2024-present

- Relevant Coursework: Advanced Algorithm, Computer System, Human Computer Integration, Generative AI, Web Development, Parallel Programming, Distributed System, etc.

York University, Lassonde School of Engineering

Toronto, ON

B.Sc. in Computer Science (Hons.) | First Class Standing With Distinction | Dean's Honour Roll

Jan.2020-Aug.2023

- Overall GPA: 8.01/9 | Major GPA: 8.29/9 | Ranking Top 1%

RESEARCH PROJECT EXPERIENCE

Building a universal python library for neuro-stimulators

Research Assistant | Supervisor: Prof. Pedro Lopes

Mar.2025-present

- Developed a universal Python library to interface with diverse neurostimulation devices, addressing fragmentation in both hardware and software ecosystems by building dedicated libraries for the NeuroStimDuino and Dummy devices.
- Focus on supporting both medical-grade neurostimulators (e.g., EMS, TMS) and custom-built devices commonly used in VR/AR and rehabilitation research.
- Design modular APIs for seamless integration with diverse hardware platforms, enabling flexible use across clinical and experimental environments.

Optimizing Data Compression Using Online Clustering for Data Migration

Research Assistant | Supervisor: Prof. Aijun An | Industry Sponsor: IBM

Oct.2022-May.2024

- Applied clustering algorithms (K-Prototypes) to large textual datasets to increase compression ratios for improving data migration performance, which was the first attempt in this field.
- Developed models for measuring data migration throughput and costs, contributing to enhanced understanding in the domain.
- Published "Optimizing Data Migration Using Online Clustering" as first author at CASCON 2023.
- Integrated offline clustering, online classification and tuple reordering into a whole data migration system. New method achieves over 24% improvement in throughput and more than 19% cost reduction compared to baseline methods.
- Co-authored "DAMOCRO: A Data Migration Framework Using Online Classification and Reordering", published in CIKM 2025.

Attentive Sensing for Long-Range Face Recognition

Research Assistant | Supervisor: Prof. James Elder; Dr. Helio Perroni Filho

Sep.2022-Apr.2023

- Aimed at designing and manufacturing a robot with long-range face recognition function
- Leveraged Ubuntu, Docker, and ROS for streamlined robotics system integration.
- Tried Haar-Cascade, HoG+SVM, and MTCNN in OpenCV for face detection to robustly locate faces under varied conditions.
- Implemented image-processing modules in PyTorch, applying a transfer-learning approach and fine-tuning a pretrained CNN (ResNet) for face recognition.
- Balanced resolution and field-of-view by merging wide-angle cameras for general detection with a high-resolution, mirror-assisted camera, enabling effective face recognition up to 35 meters away in real-world settings.

ECOs (Exotic Compact Objects) from the Dark

Research Assistant | Supervisor: Prof. Laura Sagunski (Goethe University)

Sep.2022-Apr.2023

- Collaborate with LIGO lab, which supplies data for analysis and benefits from our code contributions aimed at future dark matter core research.
- Developed an artificial neural network (FANN) to model the effects of dark matter cores on neutron stars and gravitational waves.
- Used stratified sampling to ensure diverse and representative training data, and SHAP (SHapley Additive exPlanations) analysis to interpret model predictions. These techniques yield improved predictive accuracy.

Questions Generation from Text

Research Assistant | Supervisor: Prof. Aijun An

May.2022-Jan.2023

- Managed data gathering, sorting, and analysis via Amazon Mechanical Turk, designed various interfaces, and structured data sheets.
- Utilized MTurk for efficient team collaboration and project management.

INTERNSHIP

E-Learning Internet Technology Learning Corporation

Toronto, ON

Paid Intern | Technical Department

Feb.2020-May.2021

- Led the design and development of a suite of webpages for customer registration and administration in an E-Learning APP, employing HTML, CSS, PHP, Python, JavaScript, MySQL, and Apache to create a seamless interface.
- Enhanced my backend development skills through self-study in MySQL and expanded my expertise in front-end technologies like React, React Native, and iOS App development, achieving a comprehensive grasp of the full project development cycle.

SKILLS

Programming Languages: C, C++, Java, Python, JavaScript, SQL, React, MATLAB, R, Go, HTML, CSS

Data Science Libraries: NumPy, SciPy, Panda, Scikit-Learn, TensorFlow, Scrapy, Matplotlib, Seaborn, Selenium, PyTorch

OS: Windows, Linux, Android Q, iOS, macOS, ROS

AWARDS & HONORS

- Lassonde Undergraduate Research Award (LURA) – Summer 2023
- York University Continuing Student Scholarship – Fall/Winter 2020; Fall/Winter 2021; Fall/Winter 2022
- York University International Scholarship of Merit – Summer 2020; Fall/Winter 2020